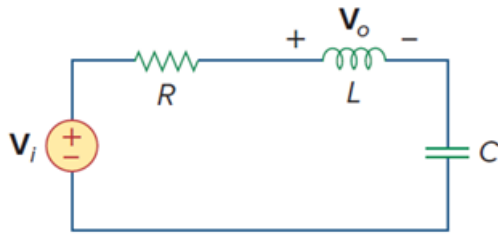


Problem

Find the transfer function $\mathbf{H}(s) = \mathbf{V_o/V_i}$ of the circuit shown in Fig. 14.71.

Figure 14.71:



Step-by-step solution

Step 1 of 2

Refer to circuit diagram in Figure 14.71 in the text book.

In s domain, inductive reactance is represented as sL , and capacitive reactance is represented as $\frac{1}{sC}$.

Calculate the voltage across inductor using voltage division rule.

$$\begin{aligned} V_o(s) &= \left(\frac{sL}{R + sL + \frac{1}{sC}} \right) V_i(s) \\ &= \left(\frac{sL}{\frac{sRC + s^2LC + 1}{sC}} \right) V_i(s) \\ &= \left(\frac{s^2LC}{sRC + s^2LC + 1} \right) V_i(s) \end{aligned}$$

Step 2 of 2

Now, calculate the transfer function.

$$\begin{aligned} \frac{V_o(s)}{V_i(s)} &= \frac{s^2LC}{sRC + s^2LC + 1} \\ &= \frac{s^2}{s^2 + \frac{R}{L}s + \frac{1}{LC}} \end{aligned}$$

Therefore, the transfer function is

$$\boxed{\frac{V_o(s)}{V_i(s)} = \frac{s^2}{s^2 + \frac{R}{L}s + \frac{1}{LC}}}$$